

Southeast University Examination Paper (in-term)

Course Name	Principles of Compiling	Examination Term	Score
Related Major	Computer & Software	Examination Form	Close test
		Test Duration	120 Mins

There are 5 problems in this paper. You can write the answers in English or Chinese on the attached paper sheets.

1. Please construct context-free grammars **with ϵ -free productions** for the following languages (20%).

(1) $\{i \mid i \in \mathbb{N}(\text{Natural number}), \text{ and } i \text{ is a palindrome, and } (i \bmod 5) = 0\}$

(2) $\{\omega \mid \omega \in (a,b,c,d)^* \text{ and the numbers of } a\text{'s}, b\text{'s} \text{ and } c\text{'s} \text{ occurred in } \omega \text{ are even, and } \omega \text{ starts with } a \text{ or } c, \text{ ends with } d\}$

2. Please construct a **DFA with minimum states** for the following regular expression. (20%)

$((a|b)^*a)^*(a|b)^*(a|b)$

3. Please **eliminate the left recursions (if there are)** and **extract maximum common left factors (if there are)** from the following context free grammar, and then decide **the resulted grammar** is whether a LL(1) grammar by **constructing the related LL(1) parsing table.**(20%)

$S \rightarrow iEtS | iEtSeS | a$

$E \rightarrow E \text{ and } F | F$

$F \rightarrow F \text{ or } G | G$

$G \rightarrow b$

4. Please **construct a LR(1) parsing table for the following ambiguous grammar with the additional conditions** that all θ_i ($i=1,2$) **has the properties of right associative law, and θ_2 has lower precedence than θ_1 .**(20%)

$E \rightarrow E \theta_1 E | E \theta_2 E | (E) | i$

5. Please show that **if a grammar G is a LL(1) grammar, then G must be a LR(1) grammar** (20%):